

## Ch. 5 — P01 (T) Trace a Single-Server Queue

Problem Bank | Chapter 5 | Type: T | Difficulty:

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A single-server barbershop operates from 9:00 AM to 1:00 PM (240 minutes). Use the inter-arrival times and service times given below. The shop is empty at time 0.

| Customer | Inter-arrival time (min) | Service time (min) |
|----------|--------------------------|--------------------|
| 1        | 0 (arrives at $t = 0$ )  | 12                 |
| 2        | 8                        | 15                 |
| 3        | 5                        | 10                 |
| 4        | 12                       | 8                  |
| 5        | 3                        | 20                 |
| 6        | 15                       | 7                  |
| 7        | 6                        | 14                 |
| 8        | 10                       | 11                 |

### Tasks

- Complete the event trace table below for all customers who *begin service* before  $t = 240$ .

| Customer | Arrival | Svc begins | Svc ends | Wait in queue | Time in system |
|----------|---------|------------|----------|---------------|----------------|
| 1        | 0       |            |          |               |                |
| 2        | 8       |            |          |               |                |
| 3        | 13      |            |          |               |                |
| 4        | 25      |            |          |               |                |
| 5        | 28      |            |          |               |                |
| 6        | 43      |            |          |               |                |
| 7        | 49      |            |          |               |                |
| 8        | 59      |            |          |               |                |

- Compute the sample mean wait in queue  $\bar{W}_q$  and mean time in system  $\bar{W}$ .

3. Estimate the time-average number in the system  $\bar{L}$  using the area-under-curve method.
4. Verify Little's Law: check whether  $\bar{L} \approx \hat{\lambda} \cdot \bar{W}$ , where  $\hat{\lambda}$  is the observed arrival rate.
5. What is the server utilization over this 240-minute period?